

Alexis SOLAR

PhD Student in AI and Telecommunications at Orange and CNAM
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Paris, France

Research Interests

LLM Agents, Multi-Agent Systems, LLM Routing, Statistical Learning, Optimization, Reinforcement Learning, Generative Models, AI for Telecom Networks (6G Core)

Education

PhD in Mathematics and Computer Science, Orange Innovation, Cedric Lab (CNAM) 2025 – Present

PhD thesis: “Adaptive LLM Agents for Future 6G Networks: Integration, Learning, and Orchestration.”

Supervisors: Stefano Secci (CNAM), Stephane Rovedakis (CNAM), Thierry Lejkin (Orange)

Master’s Degree in Mathematics and Computer Science, Sorbonne University, Paris 2023 – 2025

Relevant coursework: Machine Learning, Statistical Learning, Non-Convex and Stochastic Optimization, Computer Vision, Reinforcement Learning, Time Series, etc.

Sorbonne University High-level athlete program

Bachelor’s Degree in Mathematics, Sorbonne University, Paris 2020 – 2023

Relevant coursework: Statistics, Probability, Algebra, Linear Algebra, Differential Calculus, etc.

Sorbonne University High-level athlete program

Research Experience

PhD Researcher, Orange Innovation / CEDRIC Lab (CNAM) 2025 – Present

- Conducting research on LLM routing, with a focus on statistical learning and optimization methods for LLM routers
- Contributing to internal research and engineering efforts on multi-agent systems and LLM orchestration for 6G core network use cases.
- Participating in 6G standardization activities related to AI-native network systems and agent-based orchestration.
- Designed a proof of concept based on the Agent2Agent (A2A) Protocol to coordinate a multi-agent system for industrial telecom scenarios.
- Investigated and compared agent learning and adaptation methods, including in-context learning, RAG, fine-tuning, and reinforcement learning.

AI Research Intern, LPSM Lab (Sorbonne University) Summer 2024

Supervisor: Sylvain Le Corff

- Conducted theoretical research on diffusion models and variational inference for generative modeling.
- Collected and curated a landscape image dataset for training and evaluation.
- Implemented a U-Net architecture in Python/PyTorch and trained a diffusion model on the collected dataset.

Technical Skills

Programming: Python, SQL, C/C++, Cuda, R

ML/AI librairies: PyTorch, Scikit-Learn, Numpy, Pandas, etc.

Systems/Telecom: Networking Fundamentals, Telecom Architecture, Distributed Systems, APIs/tools Integration

Academic Activities & Service

ELLIS Summer School on AI for Research, Aalto University, Finland 2026

Reviewer: *IEEE ICC, IEEE ICNP, IEEE LATINCOM, Annals of Telecommunications, Computer Networks, IEEE Software* (2026)

Teaching Assistant: Probability and Combinatorics, Sorbonne University 2024